

Hedayet Ullah Patwary

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Summary

Final-year CSE student with a research focus on **Deep Learning**, **Computer Vision**, and **Explainable AI**. Experienced in building and evaluating ML systems for classification problems in medical and agricultural domains, with three manuscripts currently in preparation. Proficient in **Python**, **PyTorch**, and **C#/.NET**.

Education

AMERICAN INTERNATIONAL UNIVERSITY – BANGLADESH

MAY 2022 – PRESENT

- BSc in Computer Science and Engineering (Major: Information Systems). CGPA: 3.74/4

Publication

2026 IEEE 2nd International Conference on Quantum Photonics, AI, and Networking (QPAIN)

[IEEE Xplore](#)

- **Paper Title:** Evidence-Constrained Explainable Malware Detection Using Large Language Models.
- Developed the **CER framework**, integrating an **XGBoost** classifier (0.984 AUC) with **LLMs** to generate verifiable, natural-language security reports from 2,568 static features.
- Implemented a dual-layer constraint mechanism that reduced model hallucinations from 23.7% to 4.7%, ensuring 95.3% faithfulness in automated malware analysis.

Ongoing Research

Multi-Class Olive Leaf Disease Classification via Dual-Backbone CNN Fusion with 4-Method XAI

- Fusing feature representations from two complementary pre-trained CNNs for fine-grained agricultural disease classification; validating predictions through Grad-CAM, LIME, Saliency Maps, and SHAP.

Early Sepsis Prediction Using a Weighted Ensemble with Explainable AI on Non-Invasive Vital Signs

- Building a soft-voting ensemble (boosting + bagging) with leak-proof resampling for ICU sepsis onset prediction; applying KernelSHAP and LIME for global and patient-level interpretability.

Early Dyslexia Detection from Handwriting Images Using a Pre-Trained CNN

- Fine-tuning a pre-trained CNN on a large-scale handwriting dataset to classify dyslexic vs. normal writing patterns as a scalable, non-invasive screening tool.

Skills

- **Programming & Frameworks:** C#, .NET (Windows Forms, Web API), Java, PHP, Python, C++, R.
- **Data & AI:** SQL (SQL Server, MySQL), Deep Learning, Computer Vision, Explainable AI (XAI), MS Excel.
- **Web & Tools:** HTML/CSS, JavaScript, Git/GitHub, LaTeX, Visual Studio, VS Code, Kali Linux.

Academic Projects

Smart Subscription System

[GitHub](#)

- Built a full-stack **ASP.NET Core MVC (.NET 8)** platform with **3-layer architecture**, **EF Core**, **SQL Server**, **role-based access**, and an **admin dashboard** with revenue analytics.

LitVerse – Library Management System

[GitHub](#)

- Built a **Windows Forms** app in **C#** with **SQL Server** featuring role-based access, book issue/return workflows, fine calculation, and **PDF invoice generation**.

TutorSheba – Tutor Management System

[GitHub](#)

- Dual-role **Java Swing** desktop app with OOP principles, tuition job board, and **file-based data persistence**.

Network Traffic Analysis for Cybersecurity

[GitHub](#)

- Analyzed network traffic using for packet-level preprocessing, feature extraction, and EDA for **intrusion detection**.

Certifications & Poster Presentations

Cisco Networking Fundamentals – Cisco Networking Academy

[All Certificates](#)

- Certified in **routing**, **switching**, and **network security** fundamentals.

Selected Poster Presentations – AIUB

- **Science Poster:** ECO FEST 2025, Department of Economics (FASS).
- **Physics Poster:** Certificate of Achievement, Department of Physics.